

AHK-Cu

Complete Injection Guide — 50mg and 100mg Vials + 2mL BAC Water

Purity 99.8% | Endotoxin-Free | Heavy Metal-Free

AHK-Cu Vial Sizes — Side by Side



10 vials per kit | Sealed, sterile, lyophilized powder | Store 2-8 deg C refrigerated

Specification	50mg Vial	100mg Vial
Active peptide	AHK-Cu (50 mg)	AHK-Cu (100 mg)
Reconstitution vehicle	2 mL BAC Water	2 mL BAC Water
Final concentration	25 mg/mL	50 mg/mL
Vials per kit	10	10
Total kit content	500 mg	1,000 mg
Form	Lyophilized powder	Lyophilized powder
Color (reconstituted)	Clear, faint blue tint (normal)	Clear, faint blue tint (normal)

What is AHK-Cu?

AHK-Cu (Alanine-Histidine-Lysine + Copper) is a synthetic copper-binding tripeptide developed for hair-follicle stimulation, scalp microcirculation and tissue regeneration. Unlike its better-known sister peptide GHK-Cu (which targets skin and collagen broadly), AHK-Cu shows particular affinity for dermal papilla cells in hair follicles — making it a research tool for follicular research, hair density studies, and post-transplant scalp models.

Copper is delivered as a 1:1 complex with the AHK tripeptide. The copper ion (Cu²⁺) is essential for lysyl oxidase activity, antioxidant enzyme function (SOD), and the cross-linking of collagen and elastin in the follicular sheath. This explains the characteristic faint blue tint of reconstituted solution — it is the copper-peptide chromophore, NOT contamination.

Mechanism of Action

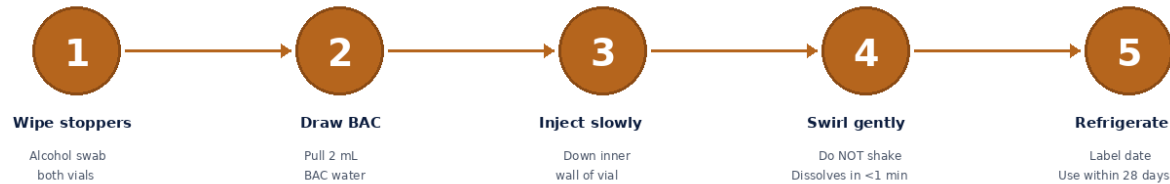
Pathway	Effect on follicle / scalp
Dermal papilla stimulation	Prolongs anagen (growth) phase; increases papilla cell proliferation in vitro
VEGF upregulation	Increases scalp microcirculation and oxygen delivery to follicles
5-alpha reductase modulation	Mild reduction in DHT formation (research data, dose-dependent)
Antioxidant activity	Cu-peptide quenches reactive oxygen species in follicular environment
Collagen / elastin synthesis	Strengthens follicular sheath and surrounding dermis
Anti-inflammatory	Downregulates TNF-alpha and IL-1 in scalp tissue models

Key Research Benefits

Domain	Reported research outcomes
Hair density	Increased terminal hair count and follicular diameter in scalp models
Hair regrowth	Re-entry of dormant follicles into anagen phase (telogen-to-anagen shift)
Post-transplant	Faster graft survival and reduced shock loss in transplant recovery models
Scalp health	Reduced sebum oxidation, improved scalp microcirculation, less itching
Wound / micro-injury	Accelerated re-epithelialization at micro-needling or scalp injection sites

Reconstitution Protocol

AHK-Cu ships as a sterile lyophilized (freeze-dried) powder. Reconstitution dissolves the peptide into a stable injectable solution. Both 50mg and 100mg vials use the same 2 mL BAC Water volume — only the final concentration changes. Always reconstitute with bacteriostatic water (BAC Water, 0.9% benzyl alcohol) for SubQ injection use.



Step	Action	Critical detail
1	Sanitize	Wipe rubber stopper of AHK-Cu vial AND BAC Water vial with fresh alcohol swab. Wait 30 seconds.
2	Draw BAC Water	Use 3 mL syringe with 21G needle. Pull back air, inject air into BAC vial, then withdraw 2.0 mL BAC Water.
3	Inject into peptide vial	Tilt AHK-Cu vial. Insert needle and let BAC Water trickle DOWN the inner glass wall. Do not squirt directly onto powder.
4	Dissolve	Roll vial gently between palms or swirl. NEVER shake. Powder dissolves in <60 sec to clear / faint blue solution.
5	Label and store	Write reconstitution date on vial. Refrigerate at 2-8 deg C. Use within 28 days.

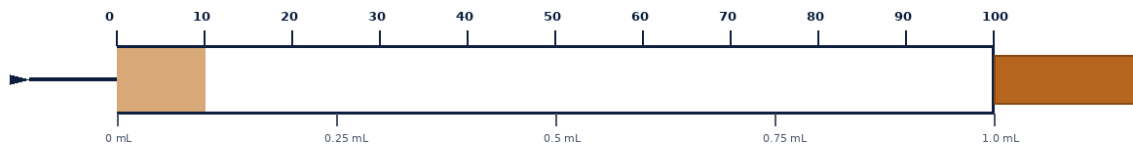
Dose Calculation

AHK-Cu is dosed in milligrams (mg). Always calculate units on an insulin syringe based on your reconstituted concentration. The formula below works for ANY peptide:

$$\text{Volume (mL)} = \text{Desired dose (mg)} \div \text{Concentration (mg/mL)}$$

Then convert to insulin units: $\text{mL} \times 100 = \text{units on a U-100 insulin syringe}$

Insulin Syringe (U-100, 1 mL)



Conversion: 100 units = 1 mL | 10 units = 0.1 mL | 1 unit = 0.01 mL

Use 0.5 mL or 1 mL insulin syringe with 29-31G needle

Dose Table — 50mg Vial (25 mg/mL after 2mL BAC)

Dose	Volume	Insulin Units	Use case
1 mg	0.04 mL	4 units	Microdose / sensitivity test
2 mg	0.08 mL	8 units	Low maintenance
3 mg	0.12 mL	12 units	Standard daily dose
4 mg	0.16 mL	16 units	Mid-range research dose
5 mg	0.20 mL	20 units	High end of typical range

Dose Table — 100mg Vial (50 mg/mL after 2mL BAC)

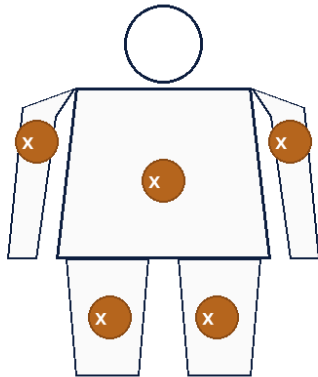
Dose	Volume	Insulin Units	Use case
1 mg	0.02 mL	2 units	Microdose
2 mg	0.04 mL	4 units	Low maintenance
3 mg	0.06 mL	6 units	Standard daily dose
4 mg	0.08 mL	8 units	Mid-range
5 mg	0.10 mL	10 units	Upper standard
7 mg	0.14 mL	14 units	Loading-phase research

Injection Technique (SubQ)

AHK-Cu is delivered via subcutaneous (SubQ) injection into the fatty layer beneath the skin. Use a U-100 insulin syringe with a 29-31G x 1/2 inch needle. The technique is identical to any other SubQ peptide.

Step	Action	Detail
1	Wash hands	Soap + warm water for 20 seconds, then dry on clean towel.
2	Prepare syringe	Draw exact dose. Tap to remove bubbles. Push out air to zero level.
3	Sanitize site	Wipe injection site with alcohol swab. Let dry 30 seconds (do not blow on it).
4	Pinch skin	Pinch 1-2 inches of skin between thumb and index finger to lift fat layer.
5	Insert at 45 deg	Quick, smooth motion. Needle fully in. No need to aspirate.
6	Inject slowly	Push plunger over 5 seconds. Wait additional 5 seconds before removing.
7	Withdraw and dispose	Pull needle out at same 45 deg angle. Release pinch. Do not rub. Place needle in sharps container.

Subcutaneous Injection Sites



Best Sites for AHK-Cu (SubQ)

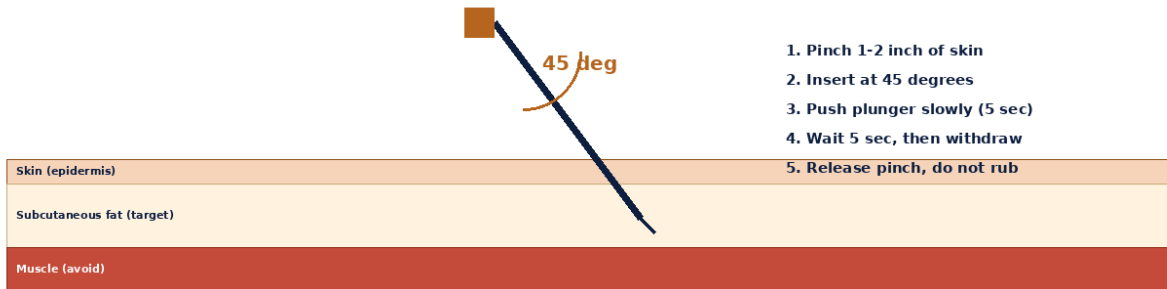
- Abdomen** **First choice**
2 inches from navel; alternate sides
- Outer thigh** **Excellent**
Upper outer quadrant; rotate L/R
- Upper arm** **Good**
Back of upper arm (triceps fat pad)
- Love handles** **Optional**
Side of waist; fattier area

AVOID: navel (2-inch radius), scars, bruises, moles, stretch marks, tattoos

Injection Angle and Site Rotation

A 45-degree angle places the needle tip in the subcutaneous fat layer — not in skin and not in muscle. For lean individuals, pinch the skin firmly to ensure adequate fat depth. For users with more abdominal fat, the angle can be 45-90 degrees safely.

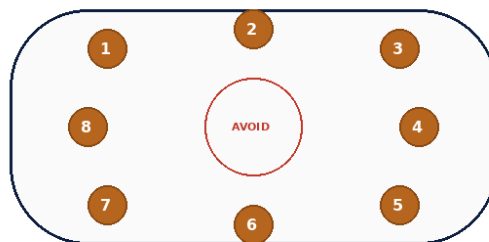
45 deg Subcutaneous Injection Angle



Site Rotation Plan

Always rotate injection sites to avoid lipohypertrophy (fatty lumps), bruising, and site fatigue. The abdomen offers 8 distinct rotation zones around the navel — never inject within 2 inches of the navel itself. Each site should rest for at least 7 days before re-use.

Site Rotation — Abdomen Map



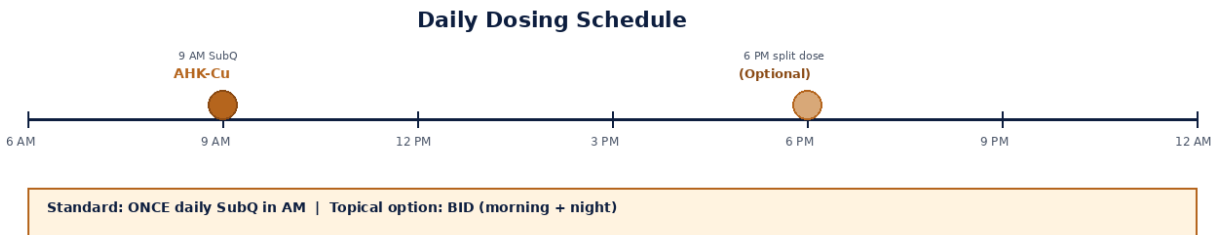
Why Rotate?

- Prevents lipohypertrophy (lumps)
- Reduces local irritation
- Maintains consistent absorption
- Each spot rests ≥ 7 days

Rotate clockwise: 1 -> 2 -> 3 -> ... -> 8 -> 1
Stay ≥ 1 inch from previous injection site

Dosing Protocol and Cycle

AHK-Cu protocols are flexible. The most common research configuration is a single daily SubQ injection into the abdomen or scalp-adjacent area, paired with optional topical application for hair-focused outcomes. Dose response plateaus around 3-5 mg/day for most subjects.



Recommended Protocols by Goal

Goal	Daily dose	Frequency	Cycle
Hair density / regrowth (research)	2-3 mg	Once daily SubQ AM	12 weeks on, 4 weeks off
Post-transplant scalp recovery	1-2 mg	Once daily SubQ	6-8 weeks continuous
Scalp microcirculation / health	1-2 mg	3-5x per week SubQ	8-12 weeks, then taper
Combined SubQ + topical	2 mg SubQ + topical	SubQ AM, topical PM	12 weeks, evaluate
Maintenance (post-cycle)	1 mg	2-3x per week	Indefinite, periodic break

Cycle Structure (12-week reference)

Phase	Weeks	Dose	Notes
Acclimation	Wk 1-2	1 mg/day	Assess tolerability, watch for redness or itching at site
Loading	Wk 3-6	2-3 mg/day	Effects begin to appear; document baseline photos
Plateau	Wk 7-10	2-3 mg/day	Steady protocol; primary research observation window
Taper	Wk 11-12	1 mg/day	Reduce gradually before break
Off-cycle	Wk 13-16	None	4-week wash-out before next cycle

Side Effects and Safety

AHK-Cu has a favorable safety profile in research settings due to its low effective dose and the natural presence of copper-binding peptides in human plasma. Most reported side effects are mild, local, and self-limiting. The following table summarizes commonly observed effects:

Effect	Frequency	Typical management
Mild redness at site	Common	Resolves in 1-2 hours; rotate sites; ensure alcohol fully dry before injection
Faint blue tint at site	Occasional	Cosmetic only — copper complex showing through skin; resolves within hours
Itching or warmth	Occasional	Cool compress; reduce dose if persistent
Bruising	Occasional	Avoid blood thinners pre-injection; use fresh needle each time
Mild fatigue	Rare	Reduce dose; assess for over-stacking with other peptides
Metallic taste	Rare	Transient, tied to copper. Lower dose if recurrent
Allergic reaction	Very rare	Discontinue immediately; seek medical advice

STOP USE IMMEDIATELY IF:

- Hives, swelling of lips/face/throat, or difficulty breathing
- Severe injection site reaction (large hard lump, persistent pain, spreading redness)
- Signs of infection at any site (warmth, pus, fever)
- Persistent metallic taste or copper-overload symptoms (nausea, headache, abdominal pain)

Storage Guide

State	Temperature	Duration	Notes
Unreconstituted powder (sealed)	2-8 deg C refrigerated	Until expiry on label	Do NOT freeze; protect from direct light; keep in original box
Reconstituted solution	2-8 deg C refrigerated	Use within 28 days	Label vial with reconstitution date; discard if cloudy or particulate appears
BAC Water (opened)	2-8 deg C refrigerated	28 days after opening	Wipe stopper with alcohol before each draw
In transit (short term)	Below 25 deg C	Up to 7 days	Use insulated cooler with ice pack; avoid direct ice contact with vial

Golden Rules — AHK-Cu

Memorize these ten rules before starting any AHK-Cu protocol. They cover the most common mistakes seen in research and self-administered settings:

01	ALWAYS use BAC Water (0.9% benzyl alcohol) — never tap, distilled, or saline.
02	Reconstitute by trickling water DOWN the inner vial wall — never squirt onto powder.
03	Swirl gently to dissolve. NEVER shake (creates foam, denatures peptide).
04	The faint blue tint is normal — it is the copper-peptide complex, not contamination.
05	Use a fresh insulin syringe + needle for every single injection.
06	Inject SubQ at 45 deg into pinched skin; avoid muscle and visible blood vessels.
07	Rotate sites in an 8-point pattern around the abdomen; rest each spot $\geq$ 7 days.
08	Start at the lowest dose (1 mg/day) for 1-2 weeks before increasing.
09	Refrigerate reconstituted vials at 2-8 deg C and discard after 28 days.
10	Stop immediately if signs of allergy, infection, or copper-overload appear.

Stack Compatibility

Stack with	Rationale	Notes
GHK-Cu	Broader skin + collagen support alongside follicular focus	Compatible; both are copper peptides — monitor for excess copper
TB-500	Adds tissue repair and wound healing capacity	Common stack; inject at separate sites
BPC-157	Anti-inflammatory + healing synergy	Excellent pairing for scalp/skin recovery
Topical minoxidil	Different mechanism (vasodilation) — additive effect	Apply at separate times of day

RESEARCH USE ONLY — IMPORTANT DISCLAIMER

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